TEST2 Phys 2049 Potential, Capacitor, DC circuits fundamentals

Name___________________________________

e = 1.60 × 10^{-19} C, $k = 1/4\pi\varepsilon_0$ = 8.99 × 10^9 N · m^2/C^2, m_{el} = 9.11 x 10^{-31} kg, ε_0 = 8.85 × 10^{-12} C^2/N m^2)

1) A negative charge (like an electron) will move 1) _______
- A) from a high potential to a low potential
- B) It never moves between points at a different pontentials
- C) from a low potential to a high potential
- D) it depends on the magnitude of the E

2) A charged particle (q = -8.0 mC), which moves in a region where the only force acting on the particle is an electric force, is released from rest at point A. At point B the kinetic energy of the particle is equal to 4.8 J. What is the electric potential difference **VB - VA**? 2) _______

A) 0.8 kV B) -0.8 kV C) -0.6 kV D) 0.6 kV

3) A +4.0-uC charge is placed on the x axis at x = +3.0 m, and a -2.0-uC charge is located on he y axis at y = -1.0 m. Point A is on the y axis at y = +4.0 m. Determine the electric potential at point A (relative to zero at infinity). 3) _______

A) 3.6 kV B) 6 kV C) 8.4 kV D) 4.8 kV

4) Consider a capacitor of capacitance C and voltage V. If you replace that capacitor with a capacitor with twice the capacitance and twice the voltage, the energy stored is increased by a factor of: 4) _______

A) 4 B) 1 C) 2 D) 8

5) Two point charges of values +3.4 uC and +6.6 uC, respectively, are separated by 0.40 m. What is the electrostatic potential energy of this 2-charge system? 5) _______

A) + 0.50 J B) -5 J C) +1 J D) -0.75 J

6) Consider a capacitor (A capacitor is comprised of two metal plates, which are separated by an insulator. one plate is positively charged and one plate is negatively charged). The separation between the 2 plates is 3cm (convert). The voltage across the plates is 12V. What is the electric field between the plates. 6) _______

A) 36V/m B) 0.360 V/m C) 4 V/m D) 400 V/m

7) If an electron is accelerated from rest through a potential difference of 9.9 kV, what is its resulting speed? 7) _______

A) 2.9 × 10^7 m/s B) 4.9 × 10^7 m/s C) 3.9 × 10^7 m/s D) 5.9 × 10^7 m/s

8) If two copper wires of the same length have different thickness, then the thicker wire has 8) _______

A) more resistance. B) less resistance. C) both the same

9) Charge of uniform linear density 3.0 nC/m is distributed along the x axis from $x = 0$ to $x = 3$ m. Which of the following integrals is correct for the electric potential (relative to zero at infinity) at the point $x = +4$ m on the x axis? 9) _______

A) $\int_0^3 \frac{9dx}{4-x}$ B) $\int_0^3 \frac{27dx}{4+x}$ C) $\int_0^3 \frac{27dx}{4-x}$ D) $\int_0^3 \frac{27dx}{x}$

10) A medium AC unit used 1,000W. It is used every night (12 hours) for 1 month (30 days). If in Miami, the rate for Kwh is $11 cents per Kwh. How much you will have to pay after one month? 10) _______

A) $10 B) $20 C) $80 D) $40

11) A 1500-W heater is connected to a 120-V line. How much heat energy does it produce in 2.0 hours? (find the energy produced) 11) _______

A) 11 MJ B) 18 MJ C) 0.18 MJ D) 1.5 kJ E) 3.0 kJ

12) A coulomb of charge flowing in a bulb filament powered by a 6-volt battery is provided with 12) _______

A) 6 newtons B) 6 amperes C) 6 watts D) 6 joules

13) The distance between the plate of a plane capacitor is increased by a factor of 10. What is true about the capacitance 13) _______

A) it increases by a factor of 10 B) it decreases by a factor of 100
C) it decreases by a factor of 10 D) It stays constant

14) A current of 5.0 A flows through an electrical device for 10 seconds. How many electrons flow through this device during this time? 14) _______

A) 2.0 E) 20 B) 0.20
C) 3.1 x 10^20 D) 31 x 10^20

15) You measure a 20.0 V potential difference across a 5.00 W resistor. What is the current flowing through it? 15) _______

A) .25 A B) 5A C) 1A D) 4A

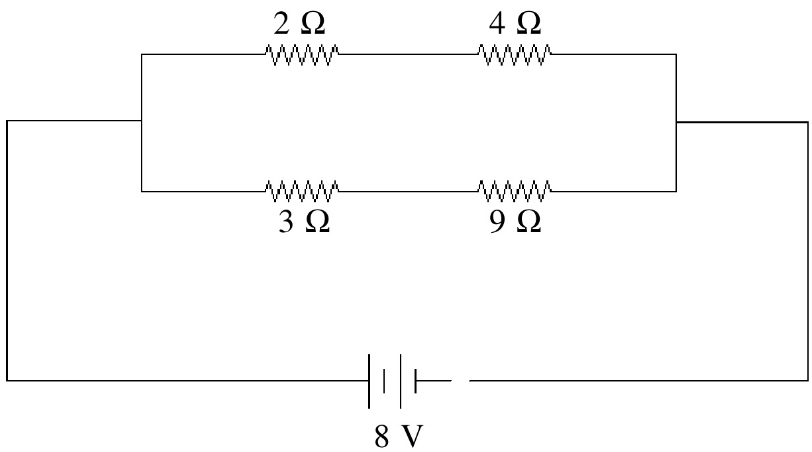
16) Equipotential surfaces around a point charge are 16) _______

A) none of them B) planar C) spherical. D) cylindrical

17) Each plate of a parallel-plate air capacitor has an area of 0.0010 m^2, and the separation of the plates is 0.050 mm. An electric field of 7.4 × 106 V/m is present between the plates. The capacitance of the capacitor, in **pF**, is closest to: 17) _______

A) 120 B) 240 C) 180 D) 360

18) Four resistors are connected across an 8-V DC battery as shown in the figure. The current through the 9-Ω resistor is closest to 18) ______



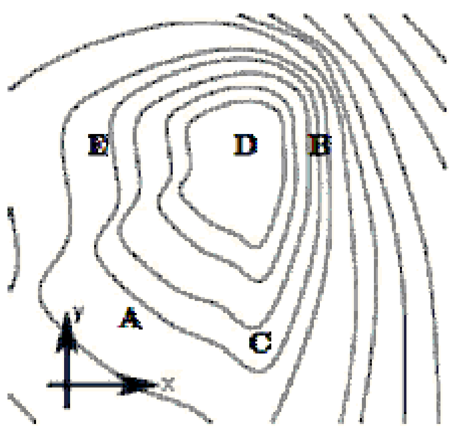
A) 0.7 A. B) 1 A. C) 0.5 A. D) 2 A. E) 0.9 A.

19) A solid conducting sphere of 10 cm radius has a net charge of 40 nC. If the potential at infinity is taken as zero, what is the potential at the center of the sphere? 19) ______

A) 3.6 ´ 104 V B) 360 uV C) 36 uV D) 3.6 ´ 103 V

20) A conductive material A and a non-conductive material B are placed in to a region permeated by an electric field E 20) ______

A) material A reinfoces E in its vicinity and E=0 inside it. material B weakens E in its visinity
B) both material B and material reinforce E in their viscinity.
C) material B reinforce E in its viscinity. material A weakens E in its viscinity E=0 inside them
D) There will be no change in the electric field configuation when ther material are placed in E.



21) 21) ______

Here are equipotentials of a distribution of charges and labelled location
Which point on the map as the largest electric field (in magnitude)

A) point B B) point A C) point C D) point D

22) 22) ____________

Electric Potential

$W = \int \vec{F} \cdot \vec{dl} = -\Delta U$

$U = \frac{kq_0 q}{r}$

$V = \frac{U}{q} = \frac{kq_0}{r}$

$\Delta V = -\int \vec{E} \cdot \vec{dl}$

$V_{cyl} = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{R}{r}\right)$

$V_{ring} = \frac{kQ}{x^2+a^2}$

$\vec{E} = -\vec{\nabla} V$

Capacitance

$C = \epsilon_0 \frac{A}{d} = \frac{Q}{V}$

$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$

$V = Ed = \frac{Qd}{\epsilon_0 A}$

$U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV$

$u = \frac{1}{2}\epsilon_0 E^2$

$C_{new} = \kappa C_{old}$

$E_{new} = \frac{E_{old}}{\kappa}$

$\varepsilon = \kappa\epsilon_0$

Answer Key
Testname: TEST2_24

1) C
2) D
3) A
4) D
5) A
6) D
7) D
8) B
9) C
10) D
11) A
12) D
13) C
14) C
15) A
16) C
17) C
18) A
19) D
20) A
21) A
22)